CRIMINAL SURVEILLANCE SYSTEM USING DEEP LEARNING FOR WEAPON DETECTION AND GESTURE ANALYSIS

A Thesis Presented To
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College of Computing and Information Sciences

In Partial Requirements for the Degree
(**BACHELOR OF SCIENCE IN COMPUTER SCIENCE**)



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ABSTRACT

TITLE: CRIMINAL SURVEILLANCE SYSTEM USING DEEP LEARNING FOR WEAPON DETECTION AND GESTURE ANALYSIS

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(RESEARCH DESIGN)

I. OBJECTIVES

- To implement YOLO and CNN algorithms for accurate real-time detection of weapons and recognition of aggressive or suspicious gestures.
- To integrate all algorithms into a unified surveillance system for efficient and reliable monitoring.
- To evaluate the system's performance by checking the accuracy, precision, recall, and percentage result of detection for both weapon and gesture recognition.

II. METHODOLOGY

III. FINDINGS

IV. RECOMMENDATIONS



KEYWORDS: Weapon & Gesture Analysis



DEDICATION

We dedicate this thesis to everyone who supported us and made this work possible. To our families, thank you for your love, care, and support through both good and difficult times. Your belief in us and even your smallest words of encouragement gave us the strength to continue.

To our friends and classmates, thank you for being with us throughout this journey. Your help, encouragement, and teamwork played a big role in our success.

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CHAPTER 1

INTRODUCTION

1.1 Project Context/Background of the Study

Crimes and violence are serious problems in society today, affecting many through theft, attacks, and other criminal activities.[1] Traditional cameras only record events and require constant human monitoring. [2][3] To reduce these risks, surveillance systems such as CCTV are often used in public places, schools, offices, and streets. However, these systems still require humans to watch them at all times.

Today's surveillance systems miss critical threats due to human operator limitations, as they can't monitor everything at once. Our autonomous system, unlike traditional designs, uses pose estimation, gesture analysis, and weapon detection.[4] Its main advantage is smart focus on human body landmarks, enabling accurate threat assessment even in crowded places.[5]

This can cause delays in responding to crimes because people can get tired or overlook details. With the help of deep learning, [6] It is now possible to create more innovative surveillance systems that can automatically detect threats such as weapons and suspicious gestures. This project, titled “Criminal Surveillance System Using Deep Learning for Weapon Detection and Gesture Analysis”, aims to provide a more reliable and efficient way of monitoring possible criminal actions.[7] It is beneficial nowadays to have peace of mind wherever we are, especially for our loved ones. Over the past twenty years, closed-circuit television (CCTV) cameras have become a key tool for crime prevention worldwide. [8] As CCTV use has spread internationally, research on its impact on crime has also increased. [9]

Recent research worldwide has used deep learning and computer vision to improve surveillance. Some studies have applied object detection models like YOLO and SSD to identify weapons in video footage. In local settings, AI is being added to security systems, but most projects are still new or only use simple motion detection. There's a notable gap in real-time systems that integrate both weapon detection and human gesture analysis to provide proactive security alerts. This leaves room for further development of innovative surveillance tools that could significantly enhance public safety and crime prevention.[10]

With modern technologies like deep learning, we can train computers to monitor video cameras and detect danger, such as someone holding a weapon or acting violently.[11] The goal is to improve safety and provide faster responses to prevent crimes through an intelligent surveillance system that identifies weapons and threatening gestures. [12]

This research is crucial because it could help make communities safer by using new developments in artificial intelligence and computer science. The goal is to create a solution that law enforcement, security agencies, and private organizations can use to respond faster and detect threats more effectively. [13] In addition, this study helps advance AI-based surveillance by working to fill the gap in unified platforms that can recognize weapons and gestures in real time. The results from this study could lay the groundwork for future improvements in innovative monitoring systems. By enabling early detection and prevention of crime and dangerous actions, the research aims to boost security and help people respond quickly to threats. [14]

In summary, the project "Criminal Surveillance System Using Deep Learning for Weapon Detection and Gesture Analysis" aims to make CCTV systems smarter by automatically detecting weapons and suspicious actions. [15]Traditional cameras require constant human monitoring, which can lead to delays. [16] By using deep learning, the system can help prevent crimes more quickly and improve safety in public areas. It also assists law enforcement by providing quick, accurate alerts on potential threats. [17]

1.2 Literature Review

Security and safety are significant issues in today's contemporary world. For a nation to be economically robust, it needs to provide a secure and safe atmosphere for both investors and tourists. Closed-circuit television (CCTV) cameras are used to monitor incidents such as thefts, but they still require human oversight and intervention.[18] Current surveillance systems overlook significant threats because human operators have limitations and cannot monitor everything simultaneously. In contrast to conventional surveillance systems, our autonomous system functions independently of human involvement by integrating pose estimation, gesture analysis, and weapon detection. The main innovation of the system is its intelligent prioritization of human body landmarks, which allows for dependable threat evaluation even in busy settings.[19]

The growing need for public safety and rapid threat response has led to the development of intelligent surveillance systems powered by artificial intelligence (AI) and deep learning. These systems are designed to detect potential criminal activities, such as the presence of weapons or threatening gestures, in real time. This section reviews previous

research and technological advancements in weapon detection, gesture recognition, and deep learning applications for surveillance.

Weapon Detection Using Deep Learning

Deep learning algorithms, especially convolutional neural networks (CNNs) and transformer-based models, have significantly improved detection accuracy in complex and dynamic environments.[20] Traditional object detection methods relied heavily on handcrafted features, which were often unreliable under variations in lighting, occlusion, and camera angles. In contrast, deep learning models automatically extract hierarchical features, enabling robust detection of concealed or partially visible weapons. [21]

Early detection of armed threats is crucial in reducing accidents and deaths resulting from armed conflicts and terrorist attacks. The most significant application of weapon detection systems would be in public areas such as airports, stadiums, central squares, and battlefields, in both urban and rural settings. [22]Modern surveillance and control systems using closed-circuit television cameras employ deep and machine learning algorithms for weapons detection, leveraging cloud architecture.[23] However, cloud computing is inefficient due to network bandwidth issues, data privacy concerns, and slow decision-making. To address these issues, edge computing can be implemented using a Raspberry Pi as an edge device (the Official model) to develop the weapons detection system.[24] To address these issues, edge computing can be applied, using a Raspberry Pi as an edge device and an efficient model to develop the weapons detection system.[25]

The image processing results are transmitted as a text report to the cloud platform for the operator's further analysis. [26]Soldiers can equip themselves with the suggested edge node and headphones for armed threat notifications, plugged into augmented reality glasses for visual data output. As a result, the application of edge computing enables ensuring data safety, increasing network bandwidth, and enabling device operation without an internet connection.[27] Thus, an independent weapon-detection system was developed that identifies weapons in 1.48 seconds without the need for an Internet connection. [28]

Gesture Recognition and Behavioral Analysis

Gesture Recognition is intended for communication between people who are hard of hearing. Communicating with others is always a difficult task for them. [29]Knowing sign language becomes a boon for those who communicate with others. Here, the sign language used is American Sign Language (ASL), which is widely used globally. Hand gestures can be a bit confusing and difficult for people who have never learned them. In this study, the user captures images of hand gestures using a webcam, and the model detects them to display the corresponding symbols. [30]

A surveillance system is needed to monitor human behavior, acting as a third eye in both crowded and uncrowded indoor and outdoor areas. This system records incidents that exhibit patterns of various human behaviors. [31]

Intelligent video monitoring is a vital security tool in public areas, using computer vision, machine learning, [32] AI analyzes video data and alerts personnel to threats. [33] These systems can recognize specific behaviors, such as loitering or unauthorized access, as

well as objects, such as persons or automobiles. Their real-time analysis capabilities have driven their growing popularity in recent years. [34]

Datasets and Benchmarks

Recent dataset efforts focused on weapon detection in surveillance contexts include the ACF (Armed CCTV Footage) dataset, which provides several thousand CCTV-style images annotated for pistols/knives and was designed to evaluate small-weapon detection on real CCTV captures [35]. The CCTV-Gun benchmark curates handgun annotations from multiple CCTV sources and provides cross-dataset evaluation protocols to better test generalization across camera configurations. Despite these additions, researchers still cite a shortage of large, diverse, and privacy-compliant CCTV corpora for weapons/violence, motivating domain adaptation and synthetic/staged data augmentation strategies. [36]

Accurate recognition of abnormal behaviors ensures public safety and security, especially in crowded scenes, but is more complicated to estimate. Closed-circuit televisions (CCTVs) installed in public places to prevent crime require automated behavior modelling mechanisms to detect abnormal activity.[37]

The developed approaches detect activities such as traffic rule violations, riots, fighting, and stampedes. Additionally, they promptly and automatically detect anomalous entities, such as abandoned luggage and weapons, at sensitive places. [38] However, identifying video anomaly methods poses several challenges due to varying environmental conditions, the ambiguous nature of anomalies, the lack of suitable datasets, and the

complex nature of human behavior. [39]In recent days, there have been only a few devoted surveys associated with deep learning related video anomaly identification, as the research domain is in its initial stages.[40]

In summary, by enabling automatic weapon and gesture detection in CCTV footage, deep learning has enhanced criminal surveillance. [41]Even in dimly lit or crowded environments, models such as YOLO, Faster R-CNN,[42] And Swin Transformer can accurately identify firearms, knives, and aggressive behaviors.[43] These systems are trained and evaluated using datasets such as ACF, CCTV-Gun, UCF-Crime, and RWF-2000. CNNs and transformers are combined in recent research to improve real-time detection.[44]

Deep learning [45] continues to improve the intelligence, speed, and dependability of surveillance systems [46] in preventing crime and maintaining public safety [47], even amid challenges such as privacy concerns and sparse datasets. The main drawbacks of these datasets include the limited range of conditions in which items are presented, the limited number of classes representing the items being detected, and the small number of instances of items from individual courses.[48] To fill this gap, a comprehensive dataset dedicated to detecting the items most frequently used in various acts of public security violations has been built. [49]. The dataset includes items such as a knife.[50], rifle, and gun, [51] which are presented in varying quality and under different environmental conditions[52].

In densely populated areas, such as schools, this can cause catastrophic damage. Therefore, the development of a rapid, accurate, and effective automatic weapon detection system is crucial for preserving social stability and saving lives.[53] [54]

The specificity of the constructed [55] The dataset provides more reliable results, giving a better idea.[56] of the effectiveness [57] of item detection in real-world conditions. [58][59][60]

1.3 Objective of the Study

The main goal of this study is to develop an intelligent surveillance system that can detect weapons and recognize violent gestures in real-time using deep learning. It aims to enhance public safety by combining weapon detection and gesture analysis into a single, efficient monitoring system.

Specifically, it aims to:

1. To implement YOLO and CNN algorithms for accurate real-time detection of weapons and recognition of aggressive or suspicious gestures.
2. To integrate all algorithms into a unified surveillance system for efficient and reliable monitoring.
3. To evaluate the system's performance by checking the accuracy, precision, recall, and percentage result of detection for both weapon and gesture recognition.

1.4 Scope and Limitations

The system will focus on detecting standard weapons, such as guns and knives, and on analyzing gestures that may suggest harmful or aggressive behavior. It will be tested using video footage in controlled settings, such as classrooms or offices.

The project will not cover all kinds of weapons or all human gestures. Also, the system may have limitations in poor lighting, crowded areas, or when the gun is hidden. This project is meant to assist security officers, not to replace them.

However, the system has some limitations. Weapons concealed beneath clothing are not detectable by it. It might not function well in crowded or dark environments. It needs a powerful computer or piece of hardware to function well in real time. Furthermore, the system may occasionally make errors, like sending alerts in the absence of a real threat.

1.5 Definition of Terms

- The **Alert System** is part of the surveillance setup that sends a warning when a threat or weapon is detected, helping notify officers or authorities to take action.
- An **algorithm** is a set of well-defined instructions in sequence to solve a problem.
- A **Surveillance System** consists of equipment and devices designed to monitor and record activities within a specific area, with the primary objective of maintaining safety and security.
- **Artificial Intelligence (AI)** is a field of technology that enables machines to perform tasks that typically require human intelligence, such as learning, reasoning, and problem-solving.

- **Computer Vision** is a technology that allows computers to see and understand visual information from cameras or videos. It is essential in this study because it helps our system to analyze what is happening in front of the camera.
- A **dataset** is a collection of images or videos used to train a computer model. In this research, the dataset includes weapons, non-weapons, and different human gestures to help the system learn what to detect.
- **Deep Learning** is a subset of Artificial Intelligence that utilizes advanced algorithms and neural networks to enable computers to identify patterns, interpret images, and recognize specific actions within data.
- **Feature extraction** involves identifying important parts of an image, such as shapes or edges. It helps the system recognize what type of object it is seeing.
- **Gesture analysis** involves detecting and interpreting body movements or hand signals to understand a person's actions or intentions.
- **Image classification** involves sorting images into different groups based on what they show. The system detects weapons or people with suspicious movements.
- **Object Detection** is a technique used to find and identify objects within an image or video, such as weapons or people performing suspicious gestures in surveillance footage.
- **Real-Time Processing** means the system analyzes data instantly as it is captured, enabling immediate detection of weapons or suspicious gestures as they occur.
- **Weapon Detection** means a system can spot dangerous items like guns or knives and alert people when they are found.

CHAPTER II



METHODOLOGY

2.1 Research Design

Experimental and Applied Design

This study employs an experimental and applied research design to develop and test a deep learning surveillance system capable of detecting weapons and analyzing human gestures in real time. The system uses YOLO for object detection and CNN for gesture classification. The goal is to build a practical surveillance tool that helps improve security and public safety through automated monitoring.

2.2 System Architecture

This section explains the technical details and structure needed to build the Criminal Surveillance System for weapon detection and gesture analysis using deep learning. It covers the key software and hardware components required for the system to function properly and connect. The design supports accurate detection, analysis, and alerting through the deep learning methods it uses.

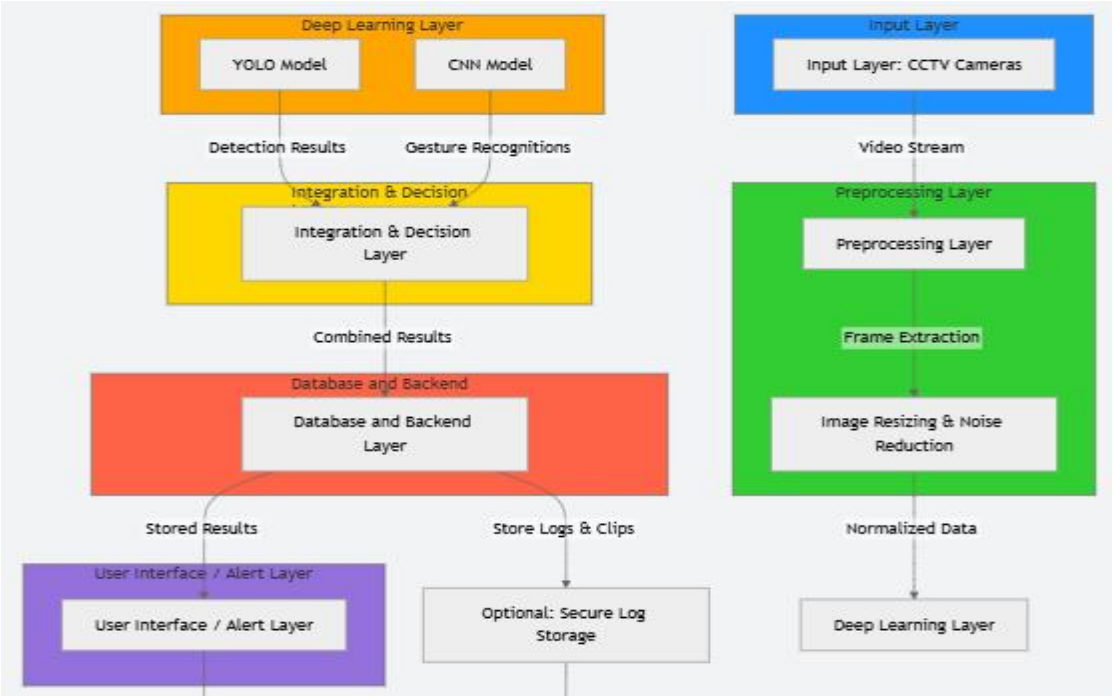


FIGURE 1: System Architecture

Figure 1 shows how the proposed system processes data from the surveillance camera. The camera records real-time video, which is then analyzed by two deep learning components.

The YOLO module identifies and monitors visible weapons, such as guns or knives, in video frames. Simultaneously, the CNN module examines human gestures and positions to detect aggressive or suspicious behaviors.

Both modules transmit their processed information to the database, where detection outcomes, timestamps, and classifications are kept. When a threat or unusual activity is identified, the system produces a warning message on the user interface to alert security staff for prompt response.

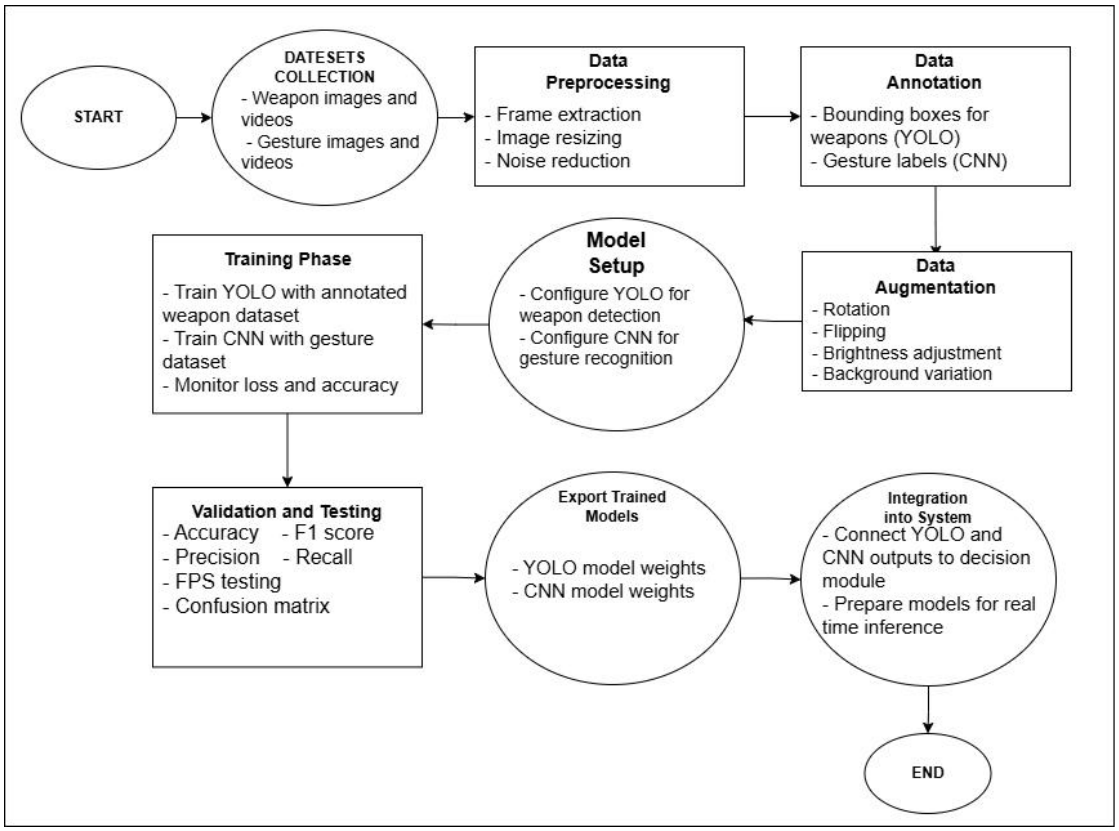


FIGURE 2: MODEL TRAINING

Figure 2 presents the step by step workflow used to train the deep learning models for weapon detection and gesture recognition. The process begins with collecting datasets that include images and videos of weapons and various human gestures. These datasets are then preprocessed through frame extraction, resizing, and noise reduction to ensure uniform quality.

After preprocessing, the data undergoes annotation, where bounding boxes are applied to weapons for YOLO and gesture labels are assigned for the CNN model. Data augmentation is then performed to increase dataset diversity by applying rotations, flips, brightness adjustments, and background variations.

Once the dataset is fully prepared, the model setup stage configures the YOLO architecture for weapon detection and the CNN architecture for gesture classification. The models are then trained using the annotated and augmented datasets. During this phase, the system learns to recognize patterns by monitoring accuracy and loss.

The trained models proceed to the validation and testing stage, where their performance is measured using metrics such as accuracy, precision, recall, F1 score, confusion matrix, and FPS evaluation. After achieving acceptable performance, the final trained models are exported and integrated into the surveillance system. This allows the system to use the trained YOLO and CNN models for real time detection once deployed.

2.3 Conceptual Framework

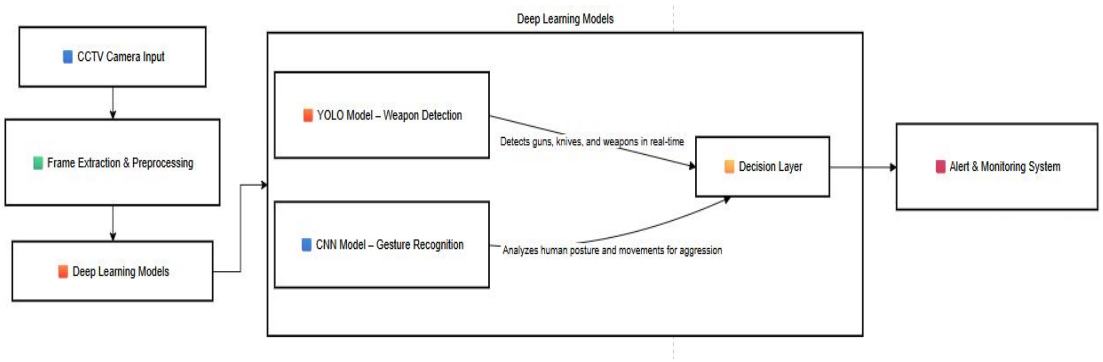


FIGURE 3: Conceptual Framework

Figure 3 illustrates the conceptual framework, detailing the flow of data from the system's inputs to its final output.

Input Phase

The process begins with the CCTV Camera Input, which continuously records live video from the monitored area. This video feed serves as the primary data source for analysis.

Process Phase

The video data is first sent to the Frame Extraction & Preprocessing module. This step prepares the raw video frames for analysis by the subsequent deep learning models.

The preprocessed frames are then fed into the Deep Learning Models, which house two central detection systems:

1. YOLO Model - Weapon Detection: This model specializes in real-time object detection, specifically to identify and track guns, knives, and weapons.
2. CNN Model - Gesture Recognition: This model specializes in image classification to analyze human posture and movements related to aggression.

The outputs from both the YOLO and CNN models are passed to the Decision Layer. This layer integrates the information from both weapon detection and gesture analysis. A threat is declared based on the combined assessment of a detected weapon, aggressive gesture, or a combination of both.

Output Phase

Once the Decision Layer identifies a potential threat or aggressive behavior, the system triggers the Alert & Monitoring System. This module is responsible for sending alerts or notifications to security personnel via a user interface, providing immediate, reliable information about a potential criminal action.

2.4 Data Gathering Procedures

The training and testing data mainly come from public video and image datasets and surveillance collections. These datasets include labeled images and videos for weapon detection and gesture analysis. YOLO is trained on datasets containing weapons such as knives and guns, while CNN is trained on datasets depicting human gestures labeled as either usual or aggressive.

All data undergo preprocessing steps such as frame extraction, annotation, and augmentation using tools like LabelImg, Roboflow, or CVAT. These steps help the model perform better and ensure a variety of lighting, camera angles, and backgrounds. All datasets follow ethical standards, and any personal information is anonymized or removed to protect

privacy. The collected data and results will be combined later for system evaluation and improvement.

2.5 Algorithms / Framework Used

YOLO (You Only Look Once)

YOLO is a real-time object detection algorithm that identifies multiple objects in an image in a single pass. It divides an image into a grid and predicts bounding boxes and class probabilities for each grid cell. In this system, YOLO is used exclusively to detect weapons, such as knives or guns, in the camera feed.

Convolutional Neural Network (CNN)

CNN is a deep learning architecture that processes visual data by learning spatial hierarchies of features. It is applied here for gesture recognition, classifying human actions as usual or aggressive. CNN analyzes body movements and patterns extracted from the video frames to determine behavioral intent.

2.6 User Interface Design

(Prototype)

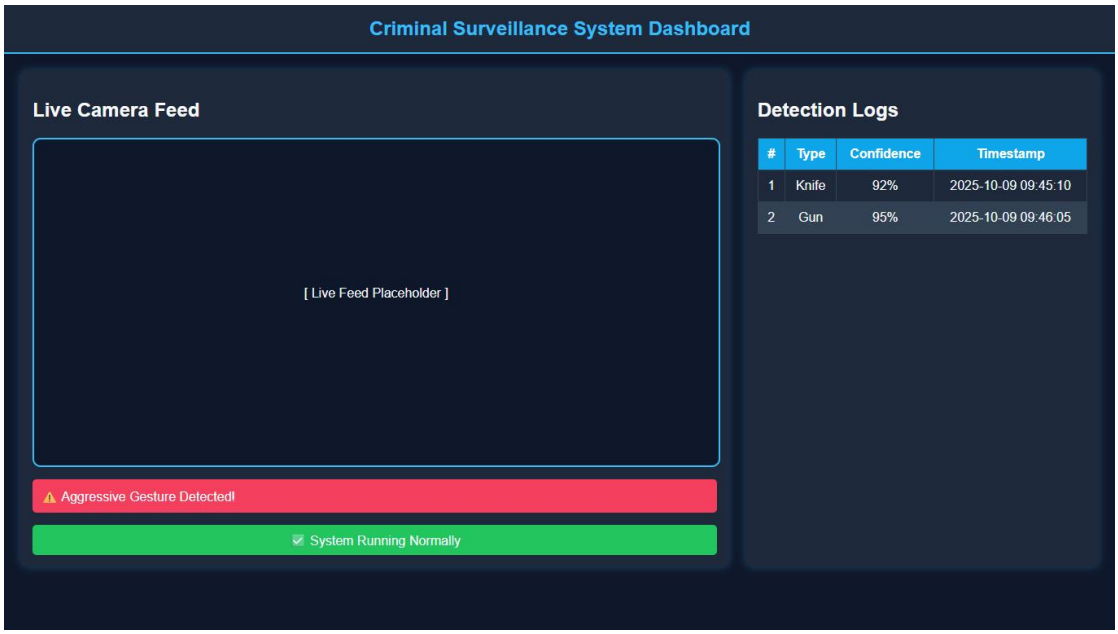


FIGURE 4: User Interface Design

Figure 4 displays the main dashboard of the simulated Criminal Surveillance System. The user interface is designed to be transparent and responsive, allowing users to view live camera feeds and detection results instantly. The left side shows the live camera feed, where users can watch the video as the system checks for weapons and gestures in real time.

The right panel shows the Detection Logs, listing detected items, such as weapons, along with their confidence levels and timestamps. This helps users track and review all system detections. Below the live feed, two indicators show the system's status: a red alert bar indicates threats or aggressive gestures, and a green bar indicates regular operation.

2.7 Software Platforms, Development Environment, and Tools

TABLE 1: Software Platforms, Development Environment, and Tools

Category	Tool / Platform	Description
Programming Language	Python	Used for algorithm implementation and integration
Deep Learning Frameworks	TensorFlow, PyTorch	For building and training YOLO and CNN models
Front-End (UI)	HTML, CSS, JavaScript, React	For building the monitoring dashboard
Back-End	Django	For handling server logic and connecting to the models
Database	SQLite	Stores detection logs and alert

		data
Development Environment	Visual Studio Code	Main environment for coding and testing
Dataset Annotation Tool	LabelImg	For labeling weapons and gestures

2.8 Hardware Requirements

TABLE 2: Hardware Requirements

Component	Specification
Processor	Intel i3 or higher
RAM	8 GB
GPU	Intel(R) UHD Graphics
Storage	500 GB SSD
Camera	HD or IP camera with 1080p resolution
Monitor	Minimum 1080p display for UI monitoring

2.8 System Evaluation/Evaluation Procedure

Evaluation Metrics

Accuracy (ACC) - Measures overall correctness of detection and classification.

Formula: ACC = (TP + TN) / (TP + TN + FP + FN)

Precision (P) - Proportion of correctly detected weapons or gestures among all detections.

Formula: P = TP / (TP + FP)

Recall (R) - Measures the ability of the system to detect all actual weapons or gestures.

Formula: R = TP / (TP + FN)

F1-Score - Harmonic mean of precision and recall.

Formula: F1 = 2 × (Precision × Recall) / (Precision + Recall)

FPS (Frames per Second) - Measures real-time processing capability.

Formula: FPS = Total Frames Processed / Total Time (in seconds)

Confusion Matrix - Evaluates model classification performance.

	Predicted Positive	Predicted Negative
Actual Positive	TP (True Positive)	FN (False Negative)
Actual Negative	FP (False Positive)	TN (True Negative)

2.9 Ethical Standard

The study is committed to following ethical standards. Before starting, the team will get written permission from the relevant authorities. Only public, simulated, or approved datasets and videos will be used. No personal information will be collected or shared at any stage.

This research addresses ethical issues related to AI, data privacy, and responsible conduct. It adheres to the institution's data security and technology standards, as well as national and international guidelines.

3.0 ETHICAL CONSIDERATIONS TEMPLATE GUIDE QUESTIONS

A. Intellectual Property Rights (IPR)

All datasets, software, and algorithms will be credited to their original creators. The team will use only proprietary data or code with permission.

B. Informed Consent

If the study uses real CCTV footage or footage of people, the team will obtain informed consent from the appropriate authorities or participants. Consent is not needed for simulated data.

C. Data Privacy and Confidentiality

All data will be stored securely and anonymized so that no one can be identified. Personal or sensitive information will not be shared. Digital files will be encrypted or protected with passwords.

D. Voluntary Participation

If volunteers take part in gesture simulation videos, their participation will be entirely voluntary. Volunteers may be research team members or others who agree to appear in simulated scenarios. Anyone involved can leave the study at any time without any adverse effects.

E. Minimization of Harm

The study does not cause any physical or psychological harm. The simulated environments and datasets used are safe and meet ethical standards.

F. Beneficence

The system is designed to enhance public safety by improving real-time threat detection and reducing societal risks.

G. Justice and Fair Participant Selection

If participants are involved, selection will be fair and unbiased. If participants are included, they will be chosen fairly and without bias. The team will not allow any data falsification or misrepresentation.

H. Data Integrity and Accuracy

The researchers will ensure that all data are collected, processed, and reported accurately and without manipulation. Data falsification or misrepresentation is strictly prohibited.

I. Transparency in Reporting

The research report will clearly explain the methods, algorithms, results, and limitations to make sure the work is transparent and can be reproduced.

J. Use of Patented or Copyrighted Materials

Copyrighted or patented materials will only be used if permission is granted or if they are open-source.

K. Ethical Considerations for Animal and Human Trials

This study does not involve animals. If any human trials are included, they will involve only simulated actions and gestures by volunteers, such as research team members or others who have agreed to participate.

L. Responsible Use of AI and Related Technologies

AI models such as YOLO and CNNs will be used responsibly, only for research and development. The team will get formal ethical approval from the institution's Research Ethics Committee before starting any experiments.

M. Ethical Approval

The team will submit an Ethical Approval Form to the institution's Research Ethics Committee. Research activities will only begin after approval is granted.

CHAPTER 3

SOFTWARE DEVELOPMENT AND TESTING

3.1 Development Process

3.2 Testing Process

- Accuracy Test

-Functionality Test

-Performance Test

CHAPTER 4

SUMMARY, CONCLUSION, RECOMMENDATION,

4.1. Summary of Findings

4.2. Conclusion

4.3. Recommendation/s

REFERENCES

[1] C. S. Sung and J. Y. Park, "Design of an intelligent video surveillance system for crime prevention: applying deep learning technology," *Multimed Tools Appl*, vol. 80, no. 26–27, pp. 34297–34309, Nov. 2021, doi: 10.1007/S11042-021-10809-Z/METRICS.

[2] R. C. Poonia, K. Upreti, P. Prabu, and K. Rajendra Prasad, "Finding Real-Time Crime Detections during Video Surveillance by Live CCTV Streaming Using the Deep Learning Models," *ACM International Conference Proceeding Series*, pp. 272–277, Apr. 2024, doi: 10.1145/3669754.3669794.

[3] T. Ruprah and H. Shrivastav, "Crime Prediction Based On Person-Weapons Relation Using Deep Learning Techniques Section A-Research Paper," *Chem. Bull*, vol. 2023, pp. 984–994, doi: 10.31838/ECB/2023.12.SI5.0118.

[4] P. Akshaya, P. Gurusai, P. B. Reddy, P. Panuganti, and A. Subhahan, "Automatic weapon detection using Deep Learning," *2023 IEEE International Conference on Research Methodologies in Knowledge Management, Artificial Intelligence and Telecommunication Engineering, RMKMATE 2023*, 2023, doi: 10.1109/RMKMATE59243.2023.10369889.

[5] M. M. Mukto *et al.*, "Design of a real-time crime monitoring system using deep learning techniques," *Intelligent Systems with Applications*, vol. 21, p. 200311, Mar. 2024, doi: 10.1016/J.ISWA.2023.200311.

[6] J. Adithya, P. Mortezaei, K. Aljawawde, M. Mohamed, R. Zgheib, and T. Mohamed, "Automated Crime Detection in CCTV Footage Using Machine Learning," *Lecture Notes on Data Engineering and Communications Technologies*, vol. 236, pp. 185–196, 2025, doi: 10.1007/978-981-96-0451-7_13.

[7] T. Santos, H. Oliveira, and A. Cunha, "Systematic review on weapon detection in surveillance footage through deep learning," *Comput Sci Rev*, vol. 51, Feb. 2024, doi: 10.1016/J.COSREV.2023.100612.

[8] "Real-Time Security Risk Assessment From CCTV Using Hand Gesture Recognition | IEEE Journals & Magazine | IEEE Xplore." Accessed: Sep. 23, 2025. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/10555026

[9] F. U. M. Ullah, M. S. Obaidat, A. Ullah, K. Muhammad, M. Hijji, and S. W. Baik, "A Comprehensive Review on Vision-Based Violence Detection in Surveillance Videos," *ACM Comput Surv*, vol. 55, no. 10, Oct. 2023, doi: 10.1145/3561971.

[10] V. Valarmathi and S. Sudha, "Enhancing public safety: a hybrid Conv_Trans-OptBiSVM approach for real-time abnormal behavior detection in crowded environments," *Signal Image Video Process*, vol. 18, no. 11, pp. 7513–7525, Nov. 2024, doi: 10.1007/S11760-024-03292-0/METRICS.

[11] H. Singh, O. Deniz, J. Ruiz-Santaquiteria, J. D. Muñoz, and G. Bueno, "DeepGun: Deep Feature-Driven One-Class Classifier for Firearm Detection Using Visual Gun Features and Human Body Pose Estimation," *Applied Sciences 2025, Vol. 15, Page 5830*, vol. 15, no. 11, p. 5830, May 2025, doi: 10.3390/APP15115830.

[12] S. Ahmed, M. T. Bhatti, M. G. Khan, B. Lövström, and M. Shahid, "Development and Optimization of Deep Learning Models for Weapon Detection in Surveillance Videos," *Applied Sciences 2022, Vol. 12, Page 5772*, vol. 12, no. 12, p. 5772, Jun. 2022, doi: 10.3390/APP12125772.

[13] R. M. Sallam, A. M. Elsayed, and N. M. Kashief, “Pose Estimation and Threat Identification: A Computer Vision Approach for Enhanced Surveillance Systems,” *Alexandria Journal of Science and Technology*, vol. 2025, no. 1, pp. 91–98, 2023, doi: 10.21608/AJST.2025.372567.1069.

[14] P. Shanthi and V. Manjula, “A systematic review on CNN-YOLO techniques for face and weapon detection in crime prevention,” *Discover Computing 2025 28:1*, vol. 28, no. 1, pp. 1–33, Sep. 2025, doi: 10.1007/S10791-025-09715-X.

[15] R. C. Poonia, K. Upreti, P. Prabu, and K. Rajendra Prasad, “Finding Real-Time Crime Detections during Video Surveillance by Live CCTV Streaming Using the Deep Learning Models,” *ACM International Conference Proceeding Series*, pp. 272–277, Apr. 2024, doi: 10.1145/3669754.3669794.

[16] “Real-Time Violence Recognition in CCTV Video Surveillance Enhanced by AI Deep Technique | IEEE Conference Publication | IEEE Xplore.” Accessed: Oct. 05, 2025. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/10883381

[17] A. Inbavalli, T. Jarshini, and M. Muralikrishnaa, “Efficient Aggressive Behaviour Detection And Alert System Employing Deep Learning Techniques,” *2024 International Conference on Automation and Computation, AUTOCOM 2024*, pp. 404–410, 2024, doi: 10.1109/AUTOCOM60220.2024.10486157.

[18] Z. Amiri, A. Heidari, N. J. Navimipour, M. Unal, and A. Mousavi, “Adventures in data analysis: a systematic review of Deep Learning techniques for pattern recognition in cyber-physical-social systems,” *Multimed Tools Appl*, vol. 83, no. 8, pp. 22909–22973, Mar. 2024, doi: 10.1007/S11042-023-16382-X/METRICS.

[19] S. T. Y. Aung and W. Kusakunniran, "A comprehensive review of gait analysis using deep learning approaches in criminal investigation," *PeerJ Comput Sci*, vol. 10, p. e2456, Nov. 2024, doi: 10.7717/PEERJ-CS.2456.

[20] G. Gupta, S. Chattopadhyay, V. Kukreja, M. Aeri, and S. Mehta, "The Arsenal Algorithm: AI-Driven Weapon Recognition with CNN -SVM Model," *2024 11th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions), ICRITO 2024*, 2024, doi: 10.1109/ICRITO61523.2024.10522391.

[21] M. M. Mukto *et al.*, "Design of a real-time crime monitoring system using deep learning techniques," *Intelligent Systems with Applications*, vol. 21, p. 200311, Mar. 2024, doi: 10.1016/J.ISWA.2023.200311.

[22] V. Kaya, S. Tuncer, and A. Baran, "Detection and Classification of Different Weapon Types Using Deep Learning," *Applied Sciences 2021, Vol. 11, Page 7535*, vol. 11, no. 16, p. 7535, Aug. 2021, doi: 10.3390/APP11167535.

[23] M. Gali, S. Dhavale, and S. Kumar, "Real-Time Image Based Weapon Detection Using YOLO Algorithms," *Communications in Computer and Information Science*, vol. 1614 CCIS, pp. 173–185, 2022, doi: 10.1007/978-3-031-12641-3_15.

[24] T. Santos, H. Oliveira, and A. Cunha, "Systematic review on weapon detection in surveillance footage through deep learning," *Comput Sci Rev*, vol. 51, p. 100612, Feb. 2024, doi: 10.1016/J.COSREV.2023.100612.

[25] T. Santos, H. Oliveira, and A. Cunha, "Systematic review on weapon detection in surveillance footage through deep learning," *Comput Sci Rev*, vol. 51, p. 100612, Feb. 2024, doi: 10.1016/J.COSREV.2023.100612.

[26] S. M. Keerthana, R. Sujitha, and P. Yazhini, "Weapon Detection for Security Using the Yolo Algorithm with Email Alert Notification," *2024 International Conference on Innovations and Challenges in Emerging Technologies, ICICET 2024*, 2024, doi: 10.1109/ICICET59348.2024.10616365.

[27] P. Wang, P. Wang, and E. Fan, "Violence detection and face recognition based on deep learning," *Pattern Recognit Lett*, vol. 142, pp. 20–24, Feb. 2021, doi: 10.1016/J.PATREC.2020.11.018.

[28] Z. R. Burnayev, D. O. Toibazarov, S. K. Atanov, H. Canbolat, Z. Y. Seitbattalov, and D. D. Kassenov, "Weapons Detection System Based on Edge Computing and Computer Vision," *International Journal of Advanced Computer Science and Applications*, vol. 14, no. 5, pp. 812–820, 2023, doi: 10.14569/IJACSA.2023.0140586.

[29] M. Koca, "Real-Time Security Risk Assessment From CCTV Using Hand Gesture Recognition," *IEEE Access*, vol. 12, pp. 84548–84555, 2024, doi: 10.1109/ACCESS.2024.3412930.

[30] V. V. Krishna Reddy, K. N. V. S. Bhuvana, K. UmaHarikka, D. Sai Teja, and J. Suguna Kumari, "Hand Gesture Recognition Using Convolutional Neural Networks and Computer Vision," *Lecture Notes in Networks and Systems*, vol. 375, pp. 583–593, 2022, doi: 10.1007/978-981-16-8763-1_48.

[31] P. Kuppusamy and V. C. Bharathi, "Human abnormal behavior detection using CNNs in crowded and uncrowded surveillance – A survey," *Measurement: Sensors*, vol. 24, p. 100510, Dec. 2022, doi: 10.1016/J.MEASEN.2022.100510.

[32] "Identifying Criminal Suspects by Human Gestures Using Deep Learning | SpringerLink." Accessed: Sep. 23, 2025. [Online]. Available: https://link.springer.com/chapter/10.1007/978-981-33-4501-0_67

[33] B. A. Omodunbi, “DEVELOPMENT OF AN INTELLIGENT SURVEILLANCE SYSTEM FOR GESTURE RECOGNITION USING MACHINE LEARNING,” *LAUTECH Journal of Engineering and Technology*, vol. 18, no. 1, pp. 173–180, May 2024, Accessed: Sep. 23, 2025. [Online]. Available: https://www.laujet.com/index.php/laujet/article/view/654

[34] B. A. Omodunbi, “DEVELOPMENT OF AN INTELLIGENT SURVEILLANCE SYSTEM FOR GESTURE RECOGNITION USING MACHINE LEARNING,” *LAUTECH Journal of Engineering and Technology*, vol. 18, no. 1, pp. 173–180, May 2024, Accessed: Oct. 08, 2025. [Online]. Available: https://www.laujet.com/index.php/laujet/article/view/654

[35] N. Hnoohom, P. Chotivatunyu, and A. Jitpattanakul, “ACF: An Armed CCTV Footage Dataset for Enhancing Weapon Detection,” *Sensors 2022, Vol. 22, Page 7158*, vol. 22, no. 19, p. 7158, Sep. 2022, doi: 10.3390/S22197158.

[36] S. Yellapragada *et al.*, “CCTV-Gun: Benchmarking Handgun Detection in CCTV Images,” Mar. 2023, doi: 10.1007/978-3-031-94895-4_18.

[37] A. Jency and K. Ramar, “A Review of Abnormal Behaviour Detection in Crowd for Video Surveillance: Advances and Trends, Datasets, Opportunities and Prospects,” *Expert Syst*, vol. 42, no. 4, p. e70013, Apr. 2025, doi: 10.1111/EXSY.70013.

[38] A. Razi *et al.*, “Deep learning serves traffic safety analysis: A forward-looking review,” *IET Intelligent Transport Systems*, vol. 17, no. 1, pp. 22–71, Jan. 2023, doi: 10.1049/ITR2.12257.

[39] P. Liu, H. L. Chi, X. Li, and J. Guo, “Effects of dataset characteristics on the performance of fatigue detection for crane operators using hybrid deep neural networks,” *Autom Constr*, vol. 132, p. 103901, Dec. 2021, doi: 10.1016/J.AUTCON.2021.103901.

[40] M. Sangeetha, M. Keerthana, B. Abirami, A. B. Maria, S. Savitha, and K. Logeswaran, “Enhancing Public Safety during Natural Disasters Using Multimodal Deep Learning Based

Analysis of Crowd-Sourced Tweets," *2nd International Conference on Artificial Intelligence and Machine Learning Applications: Healthcare and Internet of Things, AIMLA 2024*, 2024, doi: 10.1109/AIMLA59606.2024.10531594.

[41] A. S. V. Rao, S. Kainth, A. Bhattacharya, and T. Amgoth, "An efficient weapon detection system using NSGCU-DCNN classifier in surveillance," *Expert Syst Appl*, vol. 255, p. 124800, Dec. 2024, doi: 10.1016/J.ESWA.2024.124800.

[42] J. Ashok Kumar *et al.*, "Weapon Detection Identification and Classification for DCNN and YOLO-V5 Techniques," *Proceedings of 5th International Conference on IoT Based Control Networks and Intelligent Systems, ICICNIS 2024*, pp. 1317–1323, 2024, doi: 10.1109/ICICNIS64247.2024.10823365.

[43] A. Bhatt and A. Ganatra, "Deep Learning Techniques for Explosive Weapons and Arms Detection: A Comprehensive Review," *Lecture Notes in Electrical Engineering*, vol. 1078 LNEE, pp. 567–583, 2023, doi: 10.1007/978-981-99-5974-7_46.

[44] C. Sakiba, S. M. Tarannum, F. Nur, F. F. Arpan, and A. A. Anzum, "Real-time crime detection using convolutional LSTM and YOLOv7," 2023, Accessed: Sep. 24, 2025. [Online]. Available: https://dspace.bracu.ac.bd:8443/xmlui/handle/10361/22038

[45] A. Jan, G. K.-J. of I. & F. Systems, and undefined 2022, "Deep Vigilante: A deep learning network for real-world crime detection," *journals.sagepub.comA Jan, GM KhanJournal of Intelligent & Fuzzy Systems, 2022•journals.sagepub.com*, vol. 42, no. 3, pp. 1949–1961, 2022, doi: 10.3233/JIFS-211338.

[46] S. Waghchaware and R. Joshi, "Machine learning and deep learning models for human activity recognition in security and surveillance: a review," *Knowl Inf Syst*, vol. 66, no. 8, pp. 4405–4436, Aug. 2024, doi: 10.1007/S10115-024-02122-6/METRICS.

[47] E. Stephen Neal Joshua, N. Tirupati Rao, D. Bhattacharyya, and N. M. Joys, "Criminal Tendency Identification Using Deep Learning Approaches: A Novel Approach for Security Protection," *Lecture Notes in Electrical Engineering*, vol. 997 LNEE, pp. 249–258, 2023, doi: 10.1007/978-981-99-0085-5_21.

[48] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," *Proceedings - 30th IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017*, vol. 2017-January, pp. 6517–6525, Nov. 2017, doi: 10.1109/CVPR.2017.690.

[49] S. J. Alper and B. T. Karsh, "A systematic review of safety violations in industry," *Accid Anal Prev*, vol. 41, no. 4, pp. 739–754, Jul. 2009, doi: 10.1016/J.AAP.2009.03.013.

[50] M. A. Idakwo, R. E. Yoro, P. Achimugu, and O. Achimugu, "An Improved Weapons Detection and Classification System," *Journal of Network and Innovative Computing*, vol. 12, pp. 10–10, May 2024, Accessed: Oct. 11, 2025. [Online]. Available: https://cspub-jnic.org/index.php/jnic/article/view/169

[51] S. P and M. V, "Weapon detection with FMR-CNN and YOLOv8 for enhanced crime prevention and security," *Sci Rep*, vol. 15, no. 1, pp. 1–25, Dec. 2025, doi: 10.1038/S41598-025-07782-0;SUBJMETA.

[52] M. N. Naeem, O. Marma, M. Hasan, and S. Roy, "An Intelligent CCTV based Surveillance System for Realtime crimes or life threat detection," *2025 International Conference on Electrical, Computer and Communication Engineering, ECCE 2025*, 2025, doi: 10.1109/ECCE64574.2025.11013224.

[53] M. Gali, S. Dhavale, and S. Kumar, "Real-Time Image Based Weapon Detection Using YOLO Algorithms," *Communications in Computer and Information Science*, vol. 1614 CCIS, pp. 173–185, 2022, doi: 10.1007/978-3-031-12641-3_15.

[54] D. Chitravanshi, A. Malik, H. Saini, S. Avasthi, K. Agrawal, and Y. Grover, “Weapon Detection from images using YOLO and OpenCV,” *Proceedings - 2024 1st International Conference on Technological Innovations and Advance Computing, TIACOMP 2024*, pp. 560–565, 2024, doi: 10.1109/TIACOMP64125.2024.00098.

[55] A. A. Abins, P. Priyadharshini, G. C. Rohidh, and R. Cheran, “Weapon Recognition in CCTV Videos: Deep Learning Solutions for Rapid Threat Identification,” *2nd International Conference on Emerging Trends in Information Technology and Engineering, ic-ETITE 2024*, 2024, doi: 10.1109/IC-ETITE58242.2024.10493569.

[56] C. M. L. Castillo and H. A. Trapero, “Weapon Detection System on Android Devices: A Portable Machine Learning Technology,” *ACM International Conference Proceeding Series*, pp. 86–91, Jan. 2024, doi: 10.1145/3651623.3651636.

[57] N. Hnoohom, P. Chotivatunyu, N. Maitrichit, V. Sornlertlamvanich, S. Mekruksavanich, and A. Jitpattanakul, “Weapon Detection Using Faster R-CNN Inception-V2 for a CCTV Surveillance System,” *ICSEC 2021 - 25th International Computer Science and Engineering Conference*, pp. 400–405, 2021, doi: 10.1109/ICSEC53205.2021.9684649.

[58] A. S. V. Rao, S. Kainth, A. Bhattacharya, and T. Amgoth, “An efficient weapon detection system using NSGCU-DCNN classifier in surveillance,” *Expert Syst Appl*, vol. 255, p. 124800, Dec. 2024, doi: 10.1016/J.ESWA.2024.124800.

[59] M. Gali, S. Dhavale, and S. Kumar, “Real-Time Image Based Weapon Detection Using YOLO Algorithms,” *Communications in Computer and Information Science*, vol. 1614 CCIS, pp. 173–185, 2022, doi: 10.1007/978-3-031-12641-3_15.

[60] G. Wang, H. Ding, M. Duan, Y. Pu, Z. Yang, and H. Li, “Fighting against terrorism: A real-time CCTV autonomous weapons detection based on improved YOLO v4,” *Digit Signal Process*, vol. 132, p. 103790, Jan. 2023, doi: 10.1016/J.DSP.2022.103790.

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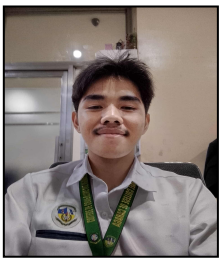
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APPENDICES

A – Letter of Approval with Certificate of Appearance and Photo
B – Instrument with Informed Consent
C – Map of the Research Locale *(Include Coordinates)*
D – Curriculum Vitae
E – Narrative & Photo Documentation
F- Ethics Committee Approval

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